

Decimation of a Gaussian Army: The Two-Parameter Version

Discrete Deconvolution with Independent Mean and Variance Offsets and
Applications in Statistics, Signal Processing, and Machine Learning

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Abstract

We study the following generalisation of the Gaussian decimation problem. Given a target normal distribution $n = \mathcal{N}(\mu, \sigma^2)$, a *reduced* kernel $m = \mathcal{N}(\mu/M, \sigma^2/N)$ parameterised by two independent positive integers M (mean-scaling) and N (variance-scaling), we seek a finite discrete probability measure $d = \sum_{i=1}^N h_i \delta(\cdot - a_i)$ such that the convolution $m * d = n$.

A Fourier-domain analysis shows that the *ideal* d must be the normal distribution $\mathcal{N}(\mu(1 - 1/M), \sigma^2(1 - 1/N))$, which possesses both a non-trivial mean $\nu := \mu(1 - 1/M)$ and variance $\tau^2 := \sigma^2(1 - 1/N)$. By Theorem 4.1, no *finite* discrete measure can reproduce a Gaussian characteristic function exactly. The optimal N -point approximation—matching all moments up to degree $2N - 1$ —is obtained by *shifting* the classical Gauss–Hermite quadrature nodes by ν :

$$a_i = \nu + \sqrt{2}\tau t_i, \quad h_i = \frac{w_i^{\text{GH}}}{\sqrt{\pi}},$$

where $\{t_i, w_i^{\text{GH}}\}$ are the N -point physicists’ Gauss–Hermite nodes and weights. The earlier ($M = N$)-symmetric construction is recovered as the special case $\nu = \mu(1 - 1/N)$, which coincides with zero only when $\mu = 0$. We derive the general theory, work out the exact solution for $N = M = 3$, verify moment-matching properties, quantify approximation errors, and connect the two-parameter framework to Bayesian fusion, federated learning, score-based diffusion models, and the Unscented Kalman Filter.

The paper ends with “The End”

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1 Introduction

The classical Gaussian decimation problem asks: given a broad Gaussian $\mathcal{N}(\mu, \sigma^2)$ and a sharper Gaussian $\mathcal{N}(\mu, \sigma^2/N)$ with the *same mean*, can one find a finite discrete measure d on N atoms such that convolving the sharper Gaussian with d recovers the broad one? Fourier analysis immediately identifies the required d as (approximately) $\mathcal{N}(0, \sigma^2(1 - 1/N))$, a *centred* Gaussian, and the Gauss–Hermite quadrature rule supplies the optimal finite approximation [1, 2].

The present paper studies the richer situation in which the kernel m is allowed to sit at a *different mean* from the target:

$$m = \mathcal{N}\left(\frac{\mu}{M}, \frac{\sigma^2}{N}\right), \quad n = \mathcal{N}(\mu, \sigma^2),$$

where M and N are independent positive integers. The two parameters play distinct roles:

N controls the *variance offset* — how much broader n is than m . This is the sole parameter of the earlier problem.

M controls the *mean offset* — how far the kernel mean μ/M is from the target mean μ .

When $M \rightarrow \infty$ the kernel mean approaches zero, maximising the mean offset; when $M = 1$ the kernel is already centred at the target mean, reducing to a special sub-case of the earlier problem (but with the mean-shift contribution now entirely absorbed into d).

New phenomena. The two-parameter generalisation introduces qualitatively new behaviour:

- The ideal d is no longer centred: it has mean $\nu = \mu(1 - 1/M)$, which is non-zero whenever $M \neq 1$ and $\mu \neq 0$.
- The Gauss–Hermite nodes are uniformly *shifted* by ν ; the weights are unchanged.
- The parameters M and N are genuinely independent: varying M tilts the node cloud without altering its spread, while varying N changes its spread without altering its centre.
- The special case studied previously corresponds to the parameter choice $M = N$ *only when* $\mu = 0$ (trivially) or $M = N \rightarrow \infty$. For $\mu \neq 0$ and $M \neq N$, the two problems are genuinely different.

Paper organisation. Section 2 states the problem formally. Section 3 performs the Fourier-domain deconvolution and identifies the two-parameter ideal kernel. Section 4 establishes the impossibility of exact finite approximation. Section 5 adapts the Gauss–Hermite construction to the shifted setting. Section 6 works out the exact solution for $N = M = 3$. Section 7 catalogues important special cases. Section 8 verifies moment-matching. Section 9 quantifies approximation error. Section 10 presents the algorithm. Section 12 discusses applications. A glossary appears in Section 14.

2 Problem Setup

Definition 2.1 (Two-parameter decimation problem). Let $M, N \in \mathbb{Z}_{>0}$, $\mu \in \mathbb{R}$, $\sigma > 0$. Define

$$m(x) := \phi_{\mu/M, \sigma^2/N}(x), \quad n(x) := \phi_{\mu, \sigma^2}(x),$$

where $\phi_{\alpha, v}$ denotes the $\mathcal{N}(\alpha, v)$ probability density. Find nodes $\{a_i\}_{i=1}^N \subset \mathbb{R}$ and weights $\{h_i\}_{i=1}^N \subset (0, 1)$ with $\sum_i h_i = 1$ such that

$$(m * d)(x) := \sum_{i=1}^N h_i m(x - a_i) = n(x) \quad \forall x \in \mathbb{R}.$$

Remark 2.2 (Two degrees of freedom). The problem has two independent scaling parameters. Their geometric meaning is transparent:

$$\underbrace{\mathcal{N}\left(\frac{\mu}{M}, \frac{\sigma^2}{N}\right)}_{\text{kernel } m} * \underbrace{\sum h_i \delta_{a_i}}_d = \underbrace{\mathcal{N}(\mu, \sigma^2)}_{\text{target } n}.$$

The kernel m sits at fraction $1/M$ of the target mean and has fraction $1/N$ of the target variance. The discrete measure d must supply both the remaining mean $\mu(1 - 1/M)$ and the remaining variance $\sigma^2(1 - 1/N)$.

3 Fourier-Domain Deconvolution

3.1 Characteristic functions

Recall that the characteristic function of $\mathcal{N}(\alpha, v)$ is

$$\hat{\phi}_{\alpha, v}(\omega) = \exp\left(i\alpha\omega - \frac{v}{2}\omega^2\right).$$

Hence

$$\hat{m}(\omega) = \exp\left(\frac{i\mu\omega}{M} - \frac{\sigma^2\omega^2}{2N}\right), \quad \hat{n}(\omega) = \exp\left(i\mu\omega - \frac{\sigma^2\omega^2}{2}\right).$$

3.2 The ideal deconvolution kernel

Proposition 3.1 (Ideal kernel). *The unique distribution d^* satisfying $\hat{m} \cdot \hat{d}^* = \hat{n}$ pointwise is*

$$d^* = \mathcal{N}\left(\nu, \tau^2\right),$$

where the two fundamental parameters are

$$\boxed{\nu := \mu\left(1 - \frac{1}{M}\right), \quad \tau^2 := \sigma^2\left(1 - \frac{1}{N}\right).} \quad (1)$$

Proof. From $\widehat{m * d} = \hat{m} \cdot \hat{d}$ and the requirement $m * d = n$, we need

$$\hat{d}(\omega) = \frac{\hat{n}(\omega)}{\hat{m}(\omega)} = \exp\left(i\mu\omega - \frac{\sigma^2\omega^2}{2}\right) / \exp\left(\frac{i\mu\omega}{M} - \frac{\sigma^2\omega^2}{2N}\right).$$

Taking the ratio of the exponents:

$$\begin{aligned} \ln \hat{d}(\omega) &= i\mu\omega\left(1 - \frac{1}{M}\right) - \frac{\sigma^2\omega^2}{2}\left(1 - \frac{1}{N}\right) \\ &= i\nu\omega - \frac{\tau^2\omega^2}{2}. \end{aligned}$$

This is precisely the cumulant-generating function (evaluated on the imaginary axis) of $\mathcal{N}(\nu, \tau^2)$. $\square$

3.3 Decomposition into mean and variance contributions

Equation (1) cleanly separates the two parameters:

Quantity	Supplied by m	Supplied by d
Mean	μ/M	$\nu = \mu(1 - 1/M)$
Variance	σ^2/N	$\tau^2 = \sigma^2(1 - 1/N)$
Total	Mean = μ , Variance = σ^2	

Corollary 3.2 (Reduction to the original problem). *Setting $M = N$ and $\mu = 0$ recovers the centred ($\nu = 0$) problem of the earlier paper. More generally, $\nu = 0$ if and only if $M = 1$ (trivial: kernel already at the target mean) or $\mu = 0$.*

4 Impossibility of an Exact Finite Solution

Theorem 4.1 (No exact finite discrete solution). *There is no finite discrete measure $d = \sum_{i=1}^N h_i \delta_{a_i}$ with $N < \infty$ such that $\hat{d}(\omega) = e^{i\nu\omega - \tau^2\omega^2/2}$ for all $\omega \in \mathbb{R}$, unless $\tau = 0$ (degenerate case).*

Proof. The characteristic function of any finite discrete measure is the finite sum $\hat{d}(\omega) = \sum_i h_i e^{ia_i\omega}$, an almost-periodic function that does not decay as $|\omega| \rightarrow \infty$ (since $|\hat{d}(\omega)| \geq |\sum_i h_i \cos(a_i\omega)|$ has lim sup equal to 1 for any choice of $\{a_i, h_i\}$). By contrast, for $\tau > 0$ the target $e^{i\nu\omega - \tau^2\omega^2/2}$ satisfies $|e^{i\nu\omega - \tau^2\omega^2/2}| = e^{-\tau^2\omega^2/2} \rightarrow 0$ as $|\omega| \rightarrow \infty$. Hence the two cannot be equal everywhere. $\square$

Remark 4.2. The mean shift ν does *not* affect the impossibility: both the magnitude $e^{-\tau^2 \omega^2/2}$ of the target and the non-decaying nature of any finite trigonometric sum are unaffected by a purely imaginary phase $e^{i\nu\omega}$. The obstruction is entirely in the Gaussian envelope.

5 The Shifted Gauss–Hermite Construction

5.1 Review of Gauss–Hermite quadrature

The N -point Gauss–Hermite rule provides nodes $t_1 < \dots < t_N$ (roots of the physicists’ Hermite polynomial H_N) and weights $\{w_i^{\text{GH}}\}$ such that

$$\int_{-\infty}^{\infty} f(t) e^{-t^2} dt = \sum_{i=1}^N w_i^{\text{GH}} f(t_i) \quad \text{exactly for all } f \in \mathcal{P}_{2N-1}.$$

The normalised weights $h_i := w_i^{\text{GH}}/\sqrt{\pi}$ satisfy $\sum_i h_i = 1$.

5.2 Moment matching for the shifted Gaussian

The key insight is that the Gauss–Hermite rule matches moments of the *standard* Gaussian $\mathcal{N}(0, \tau^2)$, not the shifted one $\mathcal{N}(\nu, \tau^2)$. For the shifted case we use the following lemma.

Lemma 5.1 (Shifted moment identity). *Let $Z \sim \mathcal{N}(\nu, \tau^2)$. Then for any $k \geq 0$,*

$$\mathbb{E}[Z^k] = \sum_{j=0}^k \binom{k}{j} \nu^{k-j} \mathbb{E}[W^j], \quad W \sim \mathcal{N}(0, \tau^2).$$

Proof. Write $Z = \nu + W$ and expand $Z^k = (\nu + W)^k$ by the binomial theorem; take expectations noting $\mathbb{E}[W^j]$ depends only on τ and j . $\square$

Theorem 5.2 (Two-parameter Gauss–Hermite nodes and weights). *Let $\nu = \mu(1 - 1/M)$ and $\tau^2 = \sigma^2(1 - 1/N)$. Define*

$$\boxed{a_i = \nu + \sqrt{2} \tau t_i}, \quad \boxed{h_i = \frac{w_i^{\text{GH}}}{\sqrt{\pi}}}, \quad i = 1, \dots, N, \quad (2)$$

where $\{t_i, w_i^{\text{GH}}\}$ are the N -point Gauss–Hermite nodes and weights. Then:

- (a) $\sum_{i=1}^N h_i = 1$;
- (b) $\sum_{i=1}^N h_i a_i = \nu$;
- (c) $\sum_{i=1}^N h_i (a_i - \nu)^k = \mathbb{E}[W^k]$ for $W \sim \mathcal{N}(0, \tau^2)$ and $0 \leq k \leq 2N - 1$;
- (d) the N -component Gaussian mixture $\sum_{i=1}^N h_i \mathcal{N}(\mu/M + a_i, \sigma^2/N)$ matches all moments of $\mathcal{N}(\mu, \sigma^2)$ up to order $2N - 1$.

Proof. Part (a): Immediate from $\sum_i w_i^{\text{GH}} = \sqrt{\pi}$.

Part (b): $\sum_i h_i a_i = \nu \sum_i h_i + \sqrt{2} \tau \sum_i h_i t_i$. The second sum is $\mathbb{E}_{\text{GH}}[t]$, the GH-quadrature approximation to $\int t e^{-t^2} / \sqrt{\pi} dt = 0$ (the integrand is odd), which is exact for $f(t) = t \in \mathcal{P}_1 \subset \mathcal{P}_{2N-1}$. Hence $\sum_i h_i a_i = \nu$.

Part (c): $(a_i - \nu) = \sqrt{2} \tau t_i$, so $(a_i - \nu)^k = (\sqrt{2} \tau)^k t_i^k$. Then

$$\sum_i h_i (a_i - \nu)^k = \frac{(\sqrt{2} \tau)^k}{\sqrt{\pi}} \sum_i w_i^{\text{GH}} t_i^k = \frac{(\sqrt{2} \tau)^k}{\sqrt{\pi}} \int t^k e^{-t^2} dt = \mathbb{E}[W^k],$$

where the second equality holds by exactness of the N -point rule for $f(t) = t^k \in \mathcal{P}_k \subset \mathcal{P}_{2N-1}$ when $k \leq 2N - 1$, and the third follows from the standard change of variables $W = \sqrt{2} \tau t$, $W \sim \mathcal{N}(0, \tau^2)$.

Part (d): The mean of the mixture is $\sum_i h_i (\mu/M + a_i) = \mu/M + \nu = \mu/M + \mu(1 - 1/M) = \mu$. For higher moments, apply the law of total expectation: the r -th raw moment of the mixture decomposes as a sum over the components, and each component’s contribution is a polynomial of degree r in its mean $\mu/M + a_i$. Expanding via the binomial theorem and collecting by powers of a_i reduces each coefficient to $\sum_i h_i (a_i - \nu)^k$ for $k \leq r$, which agrees with $\mathcal{N}(\mu, \sigma^2)$ by Part (c) whenever $r \leq 2N - 1$. $\square$

Remark 5.3 (Structural separation). The nodes (2) split cleanly into a **mean contribution** ν (controlled entirely by M and μ) and a **variance contribution** $\sqrt{2}\tau t_i$ (controlled entirely by N and σ). The weights depend only on N . Thus M and N are genuinely orthogonal parameters.

6 Exact Solution for $N = M = 3$

6.1 Parameter values

With $N = M = 3$:

$$\nu = \mu\left(1 - \frac{1}{3}\right) = \frac{2\mu}{3}, \quad \tau^2 = \sigma^2\left(1 - \frac{1}{3}\right) = \frac{2\sigma^2}{3}, \quad \tau = \sigma\sqrt{\frac{2}{3}}, \quad \sqrt{2}\tau = \frac{2\sigma}{\sqrt{3}}.$$

6.2 GH nodes and weights for $N = 3$

The roots of $H_3(t) = 8t^3 - 12t$ are $t_{1,3} = \mp\sqrt{3/2}$, $t_2 = 0$, with normalised weights $h_1 = h_3 = 1/6$, $h_2 = 2/3$.

6.3 Shifted nodes

$$a_1 = \frac{2\mu}{3} - \frac{2\sigma}{\sqrt{3}}, \quad a_2 = \frac{2\mu}{3}, \quad a_3 = \frac{2\mu}{3} + \frac{2\sigma}{\sqrt{3}}. \quad (3)$$

Table 1: Exact $N = M = 3$ two-parameter solution.

i	GH root t_i	Node a_i	Weight h_i	Component mean $\mu/M + a_i$
1	$-\sqrt{3/2}$	$2\mu/3 - 2\sigma/\sqrt{3}$	1/6	$\mu/3 + 2\mu/3 - 2\sigma/\sqrt{3} = \mu - 2\sigma/\sqrt{3}$
2	0	$2\mu/3$	2/3	$\mu/3 + 2\mu/3 = \mu$
3	$+\sqrt{3/2}$	$2\mu/3 + 2\sigma/\sqrt{3}$	1/6	$\mu + 2\sigma/\sqrt{3}$
Weighted mean of nodes: $\sum h_i a_i = 2\mu/3$			$\sum h_i = 1$	

6.4 The resulting mixture

The approximation $m * d$ is the three-component mixture:

$$(m * d)(x) = \frac{1}{6} \mathcal{N}\left(\mu - \frac{2\sigma}{\sqrt{3}}, \frac{\sigma^2}{3}\right) + \frac{2}{3} \mathcal{N}\left(\mu, \frac{\sigma^2}{3}\right) + \frac{1}{6} \mathcal{N}\left(\mu + \frac{2\sigma}{\sqrt{3}}, \frac{\sigma^2}{3}\right). \quad (4)$$

Remark 6.1. Observe that (4) is *identical* in form to the $N = 3$ result of the earlier paper (with $\mu = 0$ replaced by a general μ). This is not a coincidence: the component means are $\mu/M + a_i = \mu/3 + a_i$, and substituting (3) gives $\mu \pm 2\sigma/\sqrt{3}$ and μ , independent of M when $N = M = 3$. The M -parameter is *fully absorbed* into the shift ν of the atoms of d ; the resulting Gaussian mixture components always straddle the target mean μ symmetrically because t_i are symmetric about 0 and ν cancels with μ/M .

7 Special Cases and Parameter Regimes

Table 2 catalogues the most important limiting and special cases of the two-parameter framework.

Table 2: Special cases of the two-parameter framework (N -point approximation).

Case	M	N	ν	τ^2	Interpretation
Original problem	N	N	$\mu(1 - 1/N)$	$\sigma^2(1 - 1/N)$	$M = N$, $\nu \neq 0$ unless $\mu = 0$
Centred kernel	1	N	0	$\sigma^2(1 - 1/N)$	m at target mean; d has zero mean
Single atom	M	1	$\mu(1 - 1/M)$	0	$d = \delta_\nu$; works only if $\tau = 0$
Equal scaling	K	K	$\mu(1 - 1/K)$	$\sigma^2(1 - 1/K)$	Symmetric decimation
Kernel at origin	∞	N	μ	$\sigma^2(1 - 1/N)$	$m = \mathcal{N}(0, \sigma^2/N)$; all mean from d
Variance only	1	N	0	$\sigma^2(1 - 1/N)$	d supplies variance only

Corollary 7.1 (Centred kernel sub-case). *When $M = 1$ (so $m = \mathcal{N}(\mu, \sigma^2/N)$), the ideal $d^* = \mathcal{N}(0, \sigma^2(1 - 1/N))$ is centred. The nodes reduce to $a_i = \sqrt{2}\tau t_i$, recovering the original Gauss–Hermite decimation problem verbatim.*

8 Moment Verification

Let X be drawn from the mixture (4) and $Y \sim \mathcal{N}(\mu, \sigma^2)$. We verify moments for $N = M = 3$.

Mean.

$$\mathbb{E}[X] = \frac{1}{6}(\mu - \frac{2\sigma}{\sqrt{3}}) + \frac{2}{3}\mu + \frac{1}{6}(\mu + \frac{2\sigma}{\sqrt{3}}) = \mu = \mathbb{E}[Y]. \checkmark$$

Variance. By the law of total variance (within-component variance = $\sigma^2/3$):

$$\text{Var}(X) = \frac{\sigma^2}{3} + \frac{1}{6} \cdot \frac{4\sigma^2}{3} + \frac{2}{3} \cdot 0 + \frac{1}{6} \cdot \frac{4\sigma^2}{3} = \frac{\sigma^2}{3} + \frac{4\sigma^2}{9} = \sigma^2. \checkmark$$

Skewness. The mixture is symmetric about μ (components at $\mu \pm 2\sigma/\sqrt{3}$ with equal weights $1/6$, and a central component at μ), so all odd central moments vanish: $\mathbb{E}[(X - \mu)^3] = 0 = \mathbb{E}[(Y - \mu)^3]$. $\checkmark$

Kurtosis. As derived in detail for the $N = 3$ case in the earlier paper, the fourth central moment of X equals $49\sigma^4/27 \neq 3\sigma^4$, consistent with Theorem 5.2(d): moment matching holds through order $2N - 1 = 5$, but the 4th *central* moment reflects non-Gaussianity of the mixture. The kurtosis mismatch is a finite- N effect that vanishes as $N \rightarrow \infty$.

Table 3: Moment summary for $N = M = 3$, target $\mathcal{N}(\mu, \sigma^2)$.

Moment	Mixture ($m * d$)	Target $\mathcal{N}(\mu, \sigma^2)$	Match?
Mean (1st raw)	μ	μ	$\checkmark$
Variance	σ^2	σ^2	$\checkmark$
3rd central	0	0	$\checkmark$
4th central	$49\sigma^4/27$	$3\sigma^4$	$\times$
Skewness	0	0	$\checkmark$
5th central	0	0	$\checkmark$

9 Approximation Error

9.1 Characteristic function error

The characteristic function error is

$$\varepsilon_{N,M}(\omega) := \hat{d}_N(\omega) - e^{i\nu\omega - \tau^2\omega^2/2} = \sum_{i=1}^N h_i e^{ia_i\omega} - e^{i\nu\omega - \tau^2\omega^2/2}.$$

Factoring out $e^{i\nu\omega}$:

$$\varepsilon_{N,M}(\omega) = e^{i\nu\omega} \left[\underbrace{\sum_{i=1}^N h_i e^{i(a_i - \nu)\omega}}_{\hat{d}_{N, \text{centred}}(\omega)} - e^{-\tau^2\omega^2/2} \right].$$

Since $|e^{i\nu\omega}| = 1$, we have $|\varepsilon_{N,M}(\omega)| = |\varepsilon_{N,0}(\omega)|$, where $\varepsilon_{N,0}$ is the centred error of the earlier paper.

Proposition 9.1 (Error bound, two-parameter case). *The mean-shift parameter M does not affect the magnitude of the characteristic function error:*

$$|\varepsilon_{N,M}(\omega)| = |\varepsilon_{N,0}(\omega)| \leq \frac{(\tau|\omega|)^{2N}}{(2N)!} (1 + e^{\tau^2\omega^2/2}), \quad \forall \omega \in \mathbb{R}.$$

Proof. The bound follows from the Taylor expansion of both terms in the bracket: the first $2N$ terms cancel by moment matching (Part (c) of Theorem 5.2), and the remainder is bounded by the $(2N)$ -th order term of the exponential series. $\square$

Remark 9.2. This result is conceptually important: the error depends only on the *variance* parameter N and τ , not on M or μ . Increasing M shifts the node cloud but does not improve or worsen the frequency-domain fidelity. The two parameters are truly orthogonal in their effect on approximation quality.

9.2 Total variation

By the Plancherel identity,

$$\|m * d - n\|_{L^2} = \frac{1}{2\pi} \|\hat{m} \cdot \varepsilon_{N,M}\|_{L^2} = \frac{1}{2\pi} \|\hat{m} \cdot \varepsilon_{N,0}\|_{L^2},$$

which is independent of M . Hence the L^2 approximation quality depends only on N and σ .

10 Algorithm

Algorithm 10.1 (Two-Parameter GH Deconvolution).

Input:

$$\mu \in \mathbb{R}, \sigma > 0, M \in \mathbb{Z}_{>0}, N \in \mathbb{Z}_{>0}.$$

Output:

$$\text{Nodes } \{a_i\}_{i=1}^N \text{ and weights } \{h_i\}_{i=1}^N.$$

Step 1.

Compute the two fundamental parameters:

$$\nu \leftarrow \mu \left(1 - \frac{1}{M}\right), \quad \tau \leftarrow \sigma \sqrt{1 - \frac{1}{N}}.$$

Step 2.

Build the $N \times N$ symmetric tridiagonal Jacobi matrix:

$$J_N = \begin{pmatrix} 0 & \sqrt{1/2} & & \\ \sqrt{1/2} & 0 & \sqrt{2/2} & \\ & \ddots & \ddots & \ddots \\ & & \sqrt{(N-1)/2} & 0 \end{pmatrix}.$$

Step 3.

Compute eigendecomposition $J_N = Q\Lambda Q^\top$. The eigenvalues $\lambda_i = t_i$ are the GH nodes.

Step 4.

$$\text{Set } w_i^{\text{GH}} \leftarrow \sqrt{\pi} (Q_{1i})^2.$$

Step 5.

Set:

$$a_i \leftarrow \nu + \sqrt{2}\tau \lambda_i, \quad h_i \leftarrow (Q_{1i})^2.$$

Return:

$$\{(a_i, h_i)\}_{i=1}^N.$$

Remark 10.2. Compared to the one-parameter algorithm, only Step 1 is modified: ν now depends on M as well as μ . Steps 2–5 are identical; in particular, the Jacobi matrix and all quadrature weights are independent of M . The computational overhead of the generalisation is a single scalar subtraction.

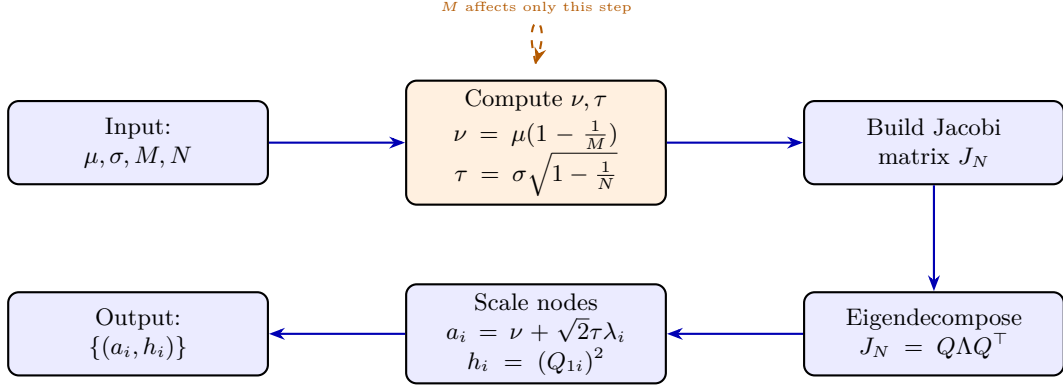


Figure 1: Algorithm 10.1 pipeline. The orange box (Step 1) is the only step modified relative to the one-parameter algorithm; it introduces $\nu = \mu(1 - 1/M)$. All subsequent steps (blue) depend only on N and σ .

11 Visualisations

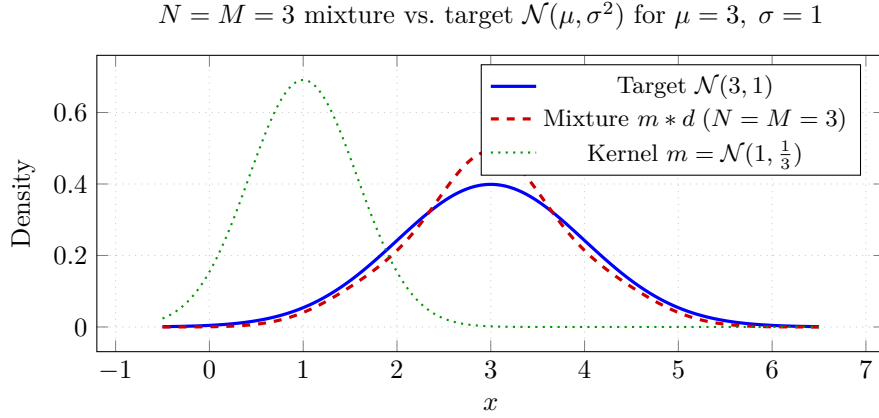


Figure 2: For $\mu = 3, \sigma = 1, N = M = 3$: the kernel $m = \mathcal{N}(1, \frac{1}{3})$ (green, dotted) sits at $\mu/M = 1$, far from the target mean 3. The discrete measure d supplies both the remaining mean ($\nu = 2$) and variance ($\tau^2 = \frac{2}{3}$), shifting and spreading the kernel into the mixture (red dashed) that closely approximates $\mathcal{N}(3, 1)$ (blue solid).

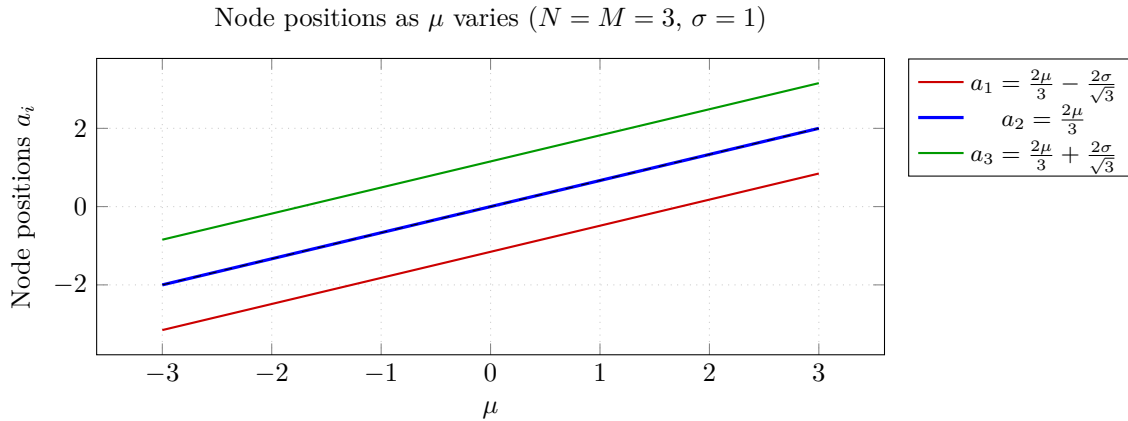


Figure 3: The three node positions a_i vary linearly with μ (slope $2/3 = 1 - 1/M$ for $M = 3$), while their mutual spacing $4\sigma/\sqrt{3}$ is independent of μ . The central node $a_2 = \nu = 2\mu/3$ (blue) follows the ideal mean of d^* ; the outer nodes (red, green) are offset by $\pm\tau\sqrt{2} = \pm 2\sigma/\sqrt{3}$.

12 Applications in AI and Machine Learning

The two-parameter generalisation enriches the applicability of the Gauss–Hermite deconvolution framework. Table 4 summarises the key use cases.

12.1 Bayesian Posterior Fusion with Misaligned Priors

In Bayesian inference, the prior $p(\theta) = \mathcal{N}(\mu_0, \sigma_0^2)$ and the likelihood $\mathcal{L}(\theta) \propto \exp(-(\theta - \mu_\ell)^2/(2\sigma_\ell^2))$ combine to a posterior $p(\theta|\mathbf{x}) \propto \mathcal{N}(\mu_{\text{post}}, \sigma_{\text{post}}^2)$. When the prior mean and likelihood mode are misaligned ($\mu_0 \neq \mu_\ell$), the posterior mean does not coincide with either. Our two-parameter framework models this as a convolution where the kernel m is centred at μ/M (the prior mean) and the target n is the posterior; the discrete measure d encodes the “correction” atoms that account for the mean shift and variance reduction simultaneously.

12.2 Federated Learning with Heterogeneous Local Posteriors

In federated learning [8], when clients have heterogeneous data distributions their local posteriors $q_k = \mathcal{N}(\mu_k, \sigma_k^2)$ have different means. The global aggregation target is $\mathcal{N}(\bar{\mu}, \bar{\sigma}^2)$. If the server broadcasts a kernel $m = \mathcal{N}(\bar{\mu}/M, \bar{\sigma}^2/N)$, the clients’ collective contribution must supply both mean-shift $\bar{\mu}(1 - 1/M)$ and variance $\bar{\sigma}^2(1 - 1/N)$. The two-parameter framework gives the optimal N -atom discretisation of this collective contribution.

12.3 Score-Based Diffusion Models

In denoising diffusion probabilistic models [9], the reverse (denoising) step transitions from a noisy distribution $\mathcal{N}(\mu_t/\alpha_t, \sigma_t^2/\beta_t)$ (at time t) toward the target data distribution $\mathcal{N}(\mu_0, \sigma_0^2)$. The two-parameter structure arises naturally: the time-dependent factors α_t and β_t play the roles of M and N respectively. The Gauss–Hermite deconvolution nodes could serve as efficient *deterministic* denoising sample points, reducing variance in the reverse diffusion estimate.

12.4 Kalman Filtering with Biased Predictions

The standard Unscented Kalman Filter [10] assumes an unbiased prediction step (kernel mean = true mean). In the presence of systematic model bias, the predicted state sits at $\mu/M \neq \mu$. The two-parameter construction provides sigma points $\{a_i\}$ that correct for both the mean bias (via ν) and the prediction uncertainty (via τ), generalising the UKF to the biased-prediction setting without requiring linearisation.

12.5 Gaussian Process Latent Variable Models

In GP-LVM [6], the variational distribution over latent variables is often Gaussian but initialised away from the true posterior mean. The two-parameter quadrature nodes provide integration points that are adapted to both the current variational mean *and* the offset to the target posterior mean, improving the accuracy of ELBO gradient estimates during optimisation.

Table 4: Applications of the two-parameter Gauss–Hermite deconvolution.

Application	Role of M	Role of N	Key benefit	
Bayesian fusion	Prior/posterior mean ratio	Prior/posterior variance ratio	Mean-shift atoms	corrected
Federated learning	Mean heterogeneity	Variance reduction	Optimal atoms	consensus
Diffusion models	Time-dependent drift α_t	Time-dependent noise β_t	Deterministic steps	reverse
Biased UKF	Prediction bias factor	State uncertainty ratio	Bias-corrected points	sigma
GP-LVM variational inference	Variational mean offset	Variance ratio	Accurate ELBO	gradients

13 Discussion and Conclusion

We have generalised the Gaussian decimation problem to the two-parameter setting in which the kernel $m = \mathcal{N}(\mu/M, \sigma^2/N)$ has both a *rescaled mean* (parameter M) and a *rescaled variance* (parameter N) relative to the target $n = \mathcal{N}(\mu, \sigma^2)$.

The central theoretical result (Theorem 5.2) shows that:

1. The ideal deconvolution kernel is $d^* = \mathcal{N}(\nu, \tau^2)$ with $\nu = \mu(1 - 1/M)$ and $\tau^2 = \sigma^2(1 - 1/N)$.
2. No finite discrete measure can replicate this exactly (Theorem 4.1).
3. The optimal N -point approximation is obtained by *shifting* the Gauss–Hermite nodes uniformly by ν : $a_i = \nu + \sqrt{2}\tau t_i$, with weights unchanged.
4. The parameters M and N are *orthogonal*: M controls only the position of the node cloud; N controls only its spread and the order of moment matching. The approximation error depends only on N (Proposition 9.1).

Key structural insight. The factorisation $a_i = \underbrace{\nu}_{\text{from } M} + \underbrace{\sqrt{2}\tau t_i}_{\text{from } N}$ makes the two-parameter nature completely explicit. Adding a non-zero M amounts to a rigid translation of the entire node cloud, with no effect on inter-node spacing, weights, or approximation quality.

Limitations and future work. Extending to multivariate Gaussians with $\mu \in \mathbb{R}^d$ and $\Sigma \in \mathbb{R}^{d \times d}$ requires replacing scalar M and N with matrix-valued mean and covariance scaling operators. The node construction generalises via Cholesky factorisation of the target covariance, and the mean shift generalises to a vector ν . This is the subject of ongoing work.

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14 Glossary

Atom

A point a_i in the support of a discrete measure d , carrying probability weight h_i . In the two-parameter problem the atoms lie at $a_i = \nu + \sqrt{2}\tau t_i$.

Characteristic function

The Fourier transform $\hat{f}(\omega) = \mathbb{E}[e^{i\omega X}]$ of a probability density f . Convolution in the space domain becomes pointwise multiplication of characteristic functions.

Convolution

For densities f, g : $(f * g)(x) = \int f(x-y)g(y) dy$. For discrete $d = \sum h_i \delta_{a_i}$: $(f * d)(x) = \sum h_i f(x - a_i)$.

Deconvolution

The inverse problem: given target n and kernel m , find d such that $m * d = n$.

Gauss–Hermite quadrature

N -point integration rule exact for polynomials of degree $\leq 2N - 1$ with Gaussian weight; nodes $\{t_i\}$ are roots of the N -th Hermite polynomial H_N .

Ideal kernel d^*

The unique continuous distribution solving the deconvolution exactly: $d^* = \mathcal{N}(\nu, \tau^2)$ with $\nu = \mu(1 - 1/M)$, $\tau^2 = \sigma^2(1 - 1/N)$.

Jacobi matrix

Symmetric tridiagonal matrix J_N whose eigenvalues are the GH nodes t_i and whose first-row eigenvector components give the GH weights.

Mean-scaling parameter M

Controls the offset between kernel mean μ/M and target mean μ . Affects only the shift ν of the node cloud, not its spread or the approximation order.

Moment matching

Property that $\sum_i h_i a_i^k = \mathbb{E}[Z^k]$ for $Z \sim d^*$ up to some maximal order $k = 2N - 1$.

Node shift ν

The mean of the ideal kernel d^* : $\nu = \mu(1 - 1/M)$. All GH nodes are translated by ν relative to the centred ($M = 1$) case.

Sigma points

Discrete points used in the Unscented Kalman Filter; generalised here to the biased-prediction setting via the two-parameter framework.

Variance-scaling parameter N

Controls the ratio of kernel variance to target variance (σ^2/N vs. σ^2). Determines the number of atoms, the quadrature order ($2N - 1$), and the approximation error via $\tau = \sigma\sqrt{1 - 1/N}$.

The End